Milestone 3.2 — Production-Ready Ad Outputs & Demo

Team Code Blooded | GigPoint Hackathon, 22 August 2026 | Problem 2: AI Ad Generation | Client: Kaladhar

Task as stated on the platform: *produce final, production-ready vertical ad outputs and a working demo of the end-to-end flow.*

Every figure, filename and image in this document was read directly from the project at build time — from `evidence/m3_delivery_manifest.json`, the run metadata and the finished export files themselves. Nothing is transcribed by hand.

1. What was produced

From **one** source product photograph, 4 videos were generated and imported, 3 passed both quality gates, and **3 final ads** were exported at 1080×1920. One video (MANSION-marble) was rejected and is **not** present in the delivered set.

Item	Value
Source image	saree_source.jpg, 500x724
Videos generated and imported	4
Rejected at quality gates	1 (MANSION-marble)
Final ads delivered	3
Delivery format	1080 x 1920, 9:16 vertical, H.264 / yuv420p, MP4
All exports conform	Yes

2. End-to-end flow demonstrated

The demo runs as one continuous workflow. Stage 4 is manual and is labelled as such throughout the product: Ekly.ai exposes no API to us, so generation is performed by hand and the run pauses until real files are brought back.

#	Stage	What it produces
1	Upload product image	Validated source image
2	Analyse product	Garment mask, colour palette, texture signature
3	Create scene briefs	6 scene briefs, each differing on >= 4 of 5 creative axes
4	Ekly handoff (MANUAL)	Operator sheet and reference image; no API available
5	Import generated videos	Each file probed; delivered format read from the file
6	Fidelity QC (Gate 1)	Per-frame product-drift score per video
7	Diversity (Gate 2)	Pairwise distance; selected set
8	Safe Cut and Export	Final 1080x1920 ads, downloadable

3. Delivery specification

Property	Delivered	Target	Met
Resolution	1080 x 1920	1080 x 1920	Yes
Aspect ratio	0.5625	0.5625 (9:16)	Yes
Orientation	Vertical	Vertical	Yes
Video codec	h264	H.264	Yes
Pixel format	yuv420p	yuv420p	Yes
Container	MP4 (brand isom)	MP4	Yes
Frame rate	30 fps	not specified	-
Duration	5.10 s	not specified	-
Exports conforming	3 of 3	all	Yes

Target placements: Instagram Reels, Instagram Stories, Facebook Reels, Meta vertical feed, YouTube Shorts.

4. Final exported ads

The three frames below are taken directly out of the delivered MP4 files at their full 1080×1920 delivery resolution — they are the shipped output, not a re-render or a preview.



BASELINE-courtyard
1080×1920 · 5.10s · 6.37 MB



MANDAP-dusk
1080×1920 · 5.10s · 6.99 MB



ROOFTOP-night
1080×1920 · 5.10s · 3.70 MB

Frame 61 of 153 shown for each (mid-clip, representative). File names as delivered: BASELINE-courtyard_1080x1920.mp4, MANDAP-dusk_1080x1920.mp4, ROOFTOP-night_1080x1920.mp4.

5. Delivery manifest (per file)

Source: `evidence/m3_delivery_manifest.json`. Values reproduced exactly.

File	Resolution	Aspect	Dur.	Frames	Size	Codec / fmt	Fit	Conforms
BASELINE-courtyard	1080x1920	0.5625	5.10s	153	6.37 MB	h264 / yuv420p	crop	Yes
MANDAP-dusk	1080x1920	0.5625	5.10s	153	6.99 MB	h264 / yuv420p	crop	Yes
ROOFTOP-night	1080x1920	0.5625	5.10s	153	3.70 MB	h264 / yuv420p	crop	Yes

All three files carry the suffix `_1080x1920.mp4` and sit in `runs/run_m2_real/exports/`.

Verification method: Resolution, aspect, duration, fps and frame count read back from the written files with probe(); codec, pixel format and container brand read with ffmpeg.

6. Format conversion: crop or pad, decided by measurement

The brief requires 9:16 vertical output for Meta placements, and this is not assumed. In every generation we ran, the Ekly 9:16 setting still returned a 768×1152 (2:3) file — all four imported videos were 2:3 on arrival. Conversion to 1080×1920 is therefore an explicit export stage, and conformance is confirmed by reading the written file back.

Converting 2:3 to 9:16 means either cropping the sides or padding them. Cropping a vertical fashion shot risks cutting the border or pallu, which would misrepresent the product; padding wastes screen area. The decision is therefore made per file from measurement: the system segments several frames and computes how much of the garment a centre crop would actually destroy, taking the worst frame.

Recorded result for all three exports:

```
crop keeps 100.00% of the garment across sampled frames (>= 99.5% required), so the full screen is used with no product loss
```

Crop removes 15.6% of frame width but 0.00% of the garment, because the model is centred. All three files therefore use **crop** and fill the full screen with no product loss.

7. Final Export screen (application)

Screenshot slot — not yet captured.

Save a screenshot of **Stage 8 — Safe Cut & export** (showing the three video players, the per-file metadata, the Download buttons and the Delivery specification panel) to:
`evidence/m3_export_screenshot.png`

then re-run `python tools/build_m3_2_pdf.py` and it will be embedded here at full width.

This slot is left for a screenshot taken on the presenting machine; it was not captured automatically.

What the Export screen provides

Requirement	Where it is shown in the application
Final 1080x1920 / 9:16 output	Delivery specification panel, and each file card
Actual generated video playback	Embedded HTML5 player on each exported ad
Duration / resolution / aspect	Per-file metadata beneath each player
Downloadable final output	Download control on each of the three ads
Suitability for vertical social	Delivery specification panel, placements listed

8. Demo procedure

The application runs locally at `http://127.0.0.1:8000` (`python -m uvicorn server.app:app --port 8000`). The reference run `run_m2_real` loads by default with all eight stages already populated, so the full flow can be walked without waiting for processing.

Step	What the audience sees
Stage 1	The single source photograph and its validation result
Stage 2	Garment isolated from background; colour and texture signature
Stage 3	Six scene briefs; axis-difference matrix proving they differ
Stage 4	Manual Ekly handoff stated plainly; prompts and reference image
Stage 5	Four imported videos, all 768x1152 - not the 9:16 that was requested
Stage 6	Per-video fidelity scores; one video rejected, with reasons
Stage 7	Pairwise distance between videos; the selected set
Stage 8	Three final ads, played in-browser, downloadable, 1080x1920

9. What we do not claim

- Generation is **not** automated. Ekly.ai exposes no API to us, so stage 4 is manual and the product says so on screen.
- **4** videos were generated, not six. Only what exists is reported.
- The four imported videos were not produced from the six briefs in stage 3; no filename matched a brief ID, and the application reports that rather than implying a link.
- Ekly wall-clock generation time has never been measured and is excluded from every timing figure in this project.
- No output in this document is a mock-up: every frame shown is decoded from a delivered MP4, and every number is read from the manifest or the run metadata.

Generated by `tools/build_m3_2_pdf.py` from `evidence/m3_delivery_manifest.json`, `runs/run_m2_real/` and the exported MP4 files.